0.1 NGSolve Notes

CoefficientFunctions: A CoefficientFunction is a function which can be evaluated on a mesh, and may be used to provide the coefficient or a right-hand-side to the variational formulation. Because typical finite element procedures iterate over elements, and map integration points from a reference element to a physical element, the evaluation of a CoefficientFunction requires a mapped integration point. The mapped integration point contains the coordinate, as well as the Jacobian, but also element number and region index. This allows the efficient evaluation of a wide class of functions needed for finite element computations.

TestFunction and TrialFunction: TestFunction and TrialFunction are symbolic objects used to define the variational (weak) form of a partial differential equation. They act as placeholders for the unknown solution and the weighting function during the construction of bilinear and linear forms, allowing you to write equations in a natural mathematical syntax. **Key Concepts and Usage**

- **Definition.** They are created from a defined space (`fes`), typically using `u = fes.TrialFunction()` and `v = fes.TestFunction()`, or simultaneously with `u, v = fes.TnT()`.
- **Symbolic operation.** They are used to symbolically express bilinear forms (e.g., $a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx$) before assembly.
- **Spaces.** Test and trial functions belong to a `FESpace` (e.g. H^1, L^2, $H(\mathrm{div})$) that defines their continuity and polynomial order.

GridFunction: A GridFunction is a function defined on a mesh that can be used to store and manipulate the solution of a partial differential equation. It is a container for the coefficient vector of a finite element space, and can be used to store the solution of a partial differential equation.